TASK 1 – Conditional Statements

1. Aim:-

To create a Python program that takes a student's marks as input and displays the grade using if, elif, and else statements.

2. Python Code:-

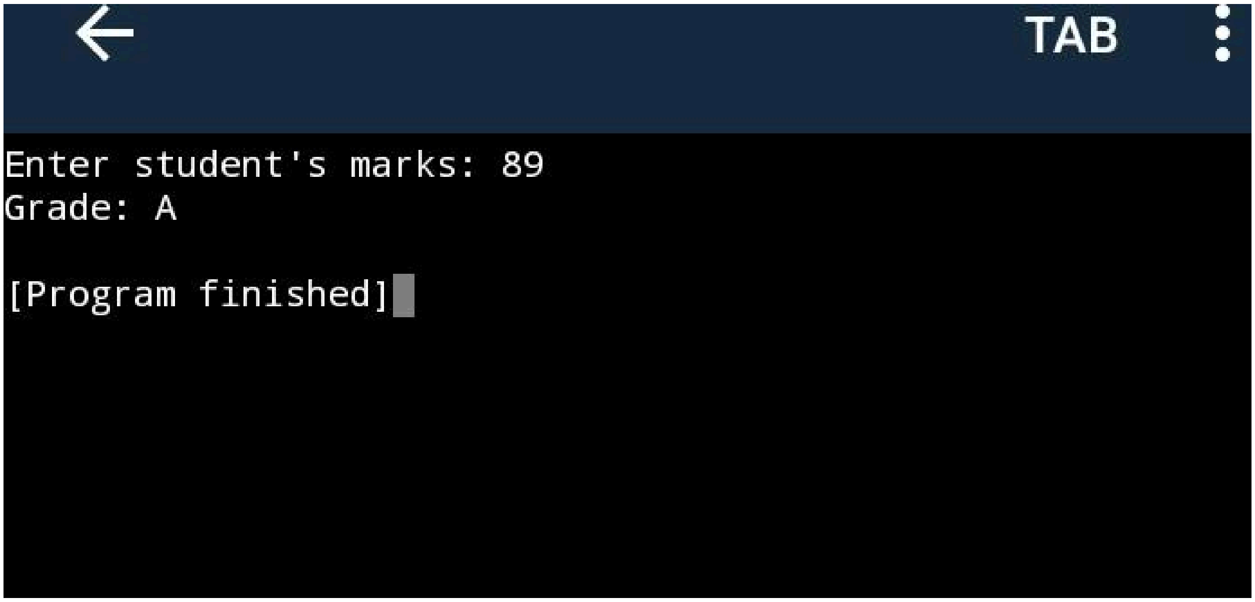
Task 1: Student Grade Calculator

new* new* new* new* new*

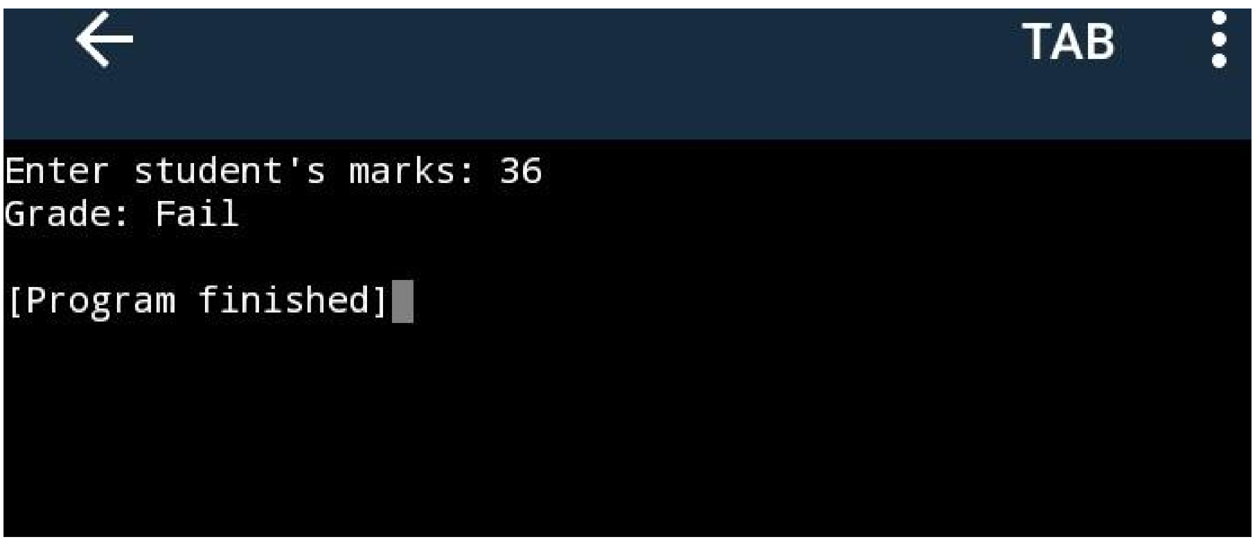
```python

marks = float(input("Enter student's marks: "))

if marks < 0 or marks > 100:
    print("Invalid marks! Please enter mark between 0 and 100.")

elif marks >= 90:
    print("Grade: A+")

elif marks >= 80:
    print("Grade: A")

elif marks >= 70:
    print("Grade: B")

elif marks >= 60:
    print("Grade: C")

elif marks >= 50:
    print("Grade: D")

else:
    print("Grade: Fail")
```

3. Output:-

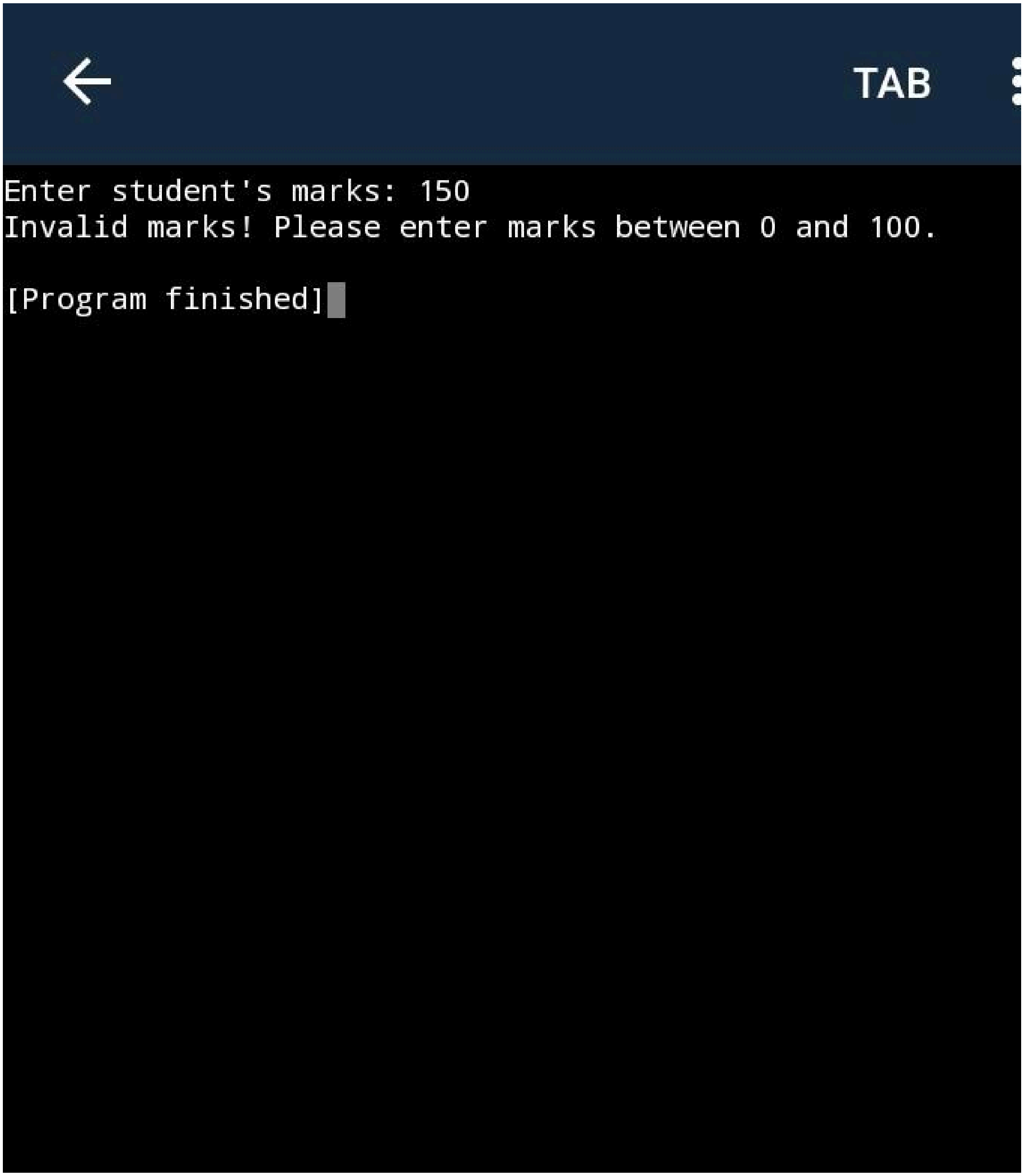
#1)


TAB
Enter student's marks: 89
Grade: A
[Program finished]

#2)


TAB
Enter student's marks: 36
Grade: Fail
[Program finished]

#3)



Explanation:
This program accepts the student's marks as input and uses if, elif, and else conditional statements to determine the grade. It also checks whether the entered marks are valid. Marks should be between 0 and 100.

TASK 2 – Loops:-

1. Aim:-

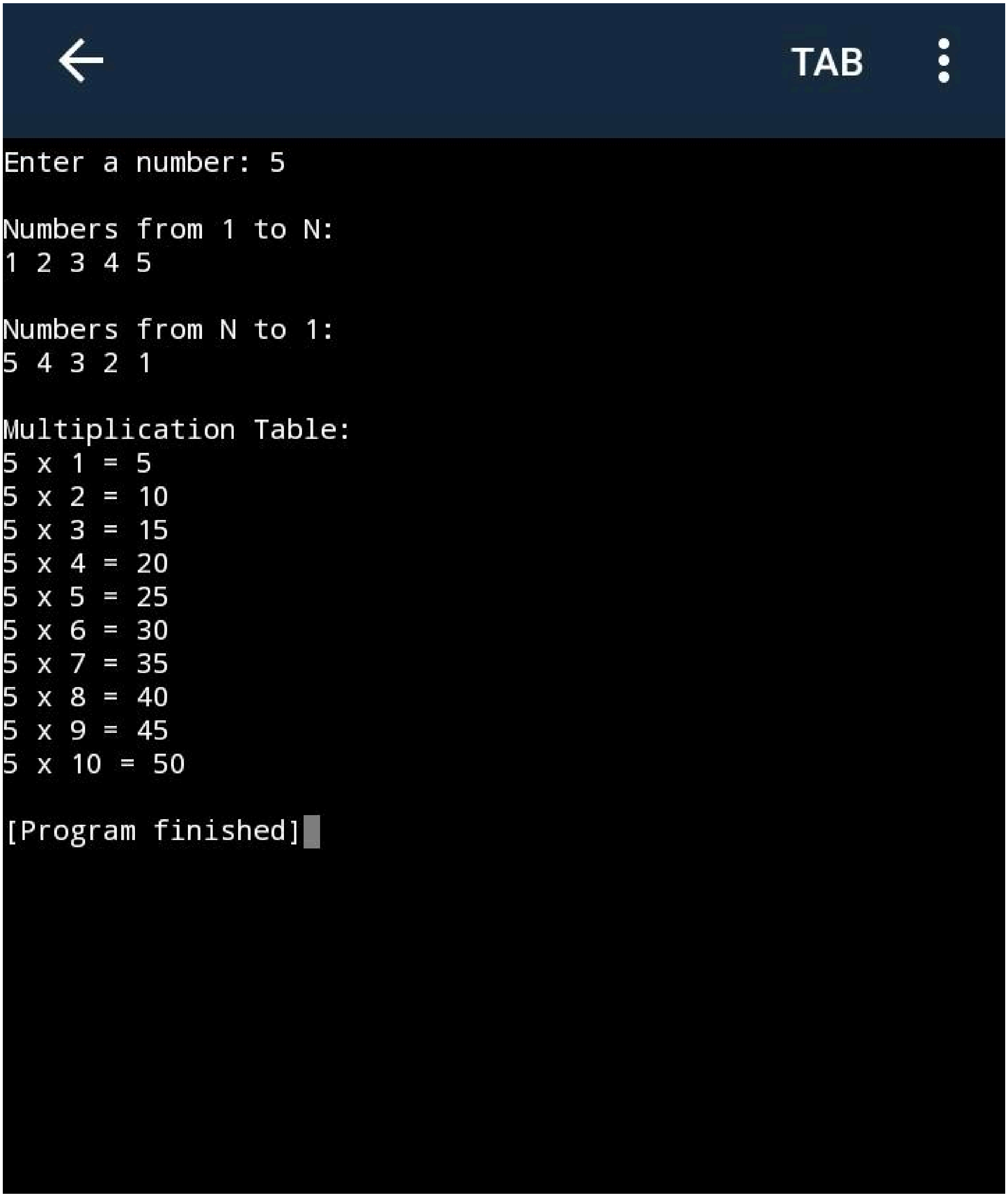
To create a Python program using loops to display numbers from 1 to N, N to 1, and a multiplication table.

2. Python Code:-

new* new* new* new* new*

```python

n = int(input("Enter a number: "))

print("\nNumbers from 1 to N:")
for i in range(1, n + 1):
    print(i, end=" ")

print("\n\nNumbers from N to 1:")
for i in range(n, 0, -1):
    print(i, end=" ")

print("\n\nMultiplication Table:")
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Output:-




```
Enter a number: 5

Numbers from 1 to N:
1 2 3 4 5

Numbers from N to 1:
5 4 3 2 1

Multiplication Table:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

[Program finished]
```

Explanation:

This program uses a for loop to print numbers from 1 to N, another for loop to print numbers from N to 1, and a third loop to display the multiplication table from 1 to 10.

TASK 3 – Functions

1. Aim:-
To create a simple calculator using user-defined functions for addition, subtraction, multiplication, division, and modulus.

2. Python Code:-

new* new* new* new* new*

```python
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    return a / b

def modulus(a, b):
    return a % b


a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("\nAddition:", add(a, b))
print("Subtraction:", subtract(a, b))
print("Multiplication:", multiply(a, b))

if b != 0:
    print("Division:", divide(a, b))
    print("Modulus:", modulus(a, b))
```

Output:-

```
Enter first number: 20
Enter second number: 5

Addition: 25.0
Subtraction: 15.0
Multiplication: 100.0
Division: 4.0
Modulus: 0.0

[Program finished]
```

Explanation:

This program uses five user-defined functions: add(), subtract(), multiply(), divide(), and modulus(). Each function performs a different arithmetic operation. The program takes two numbers from the user and displays the results.

TASK 4 – Scope of Variables:-

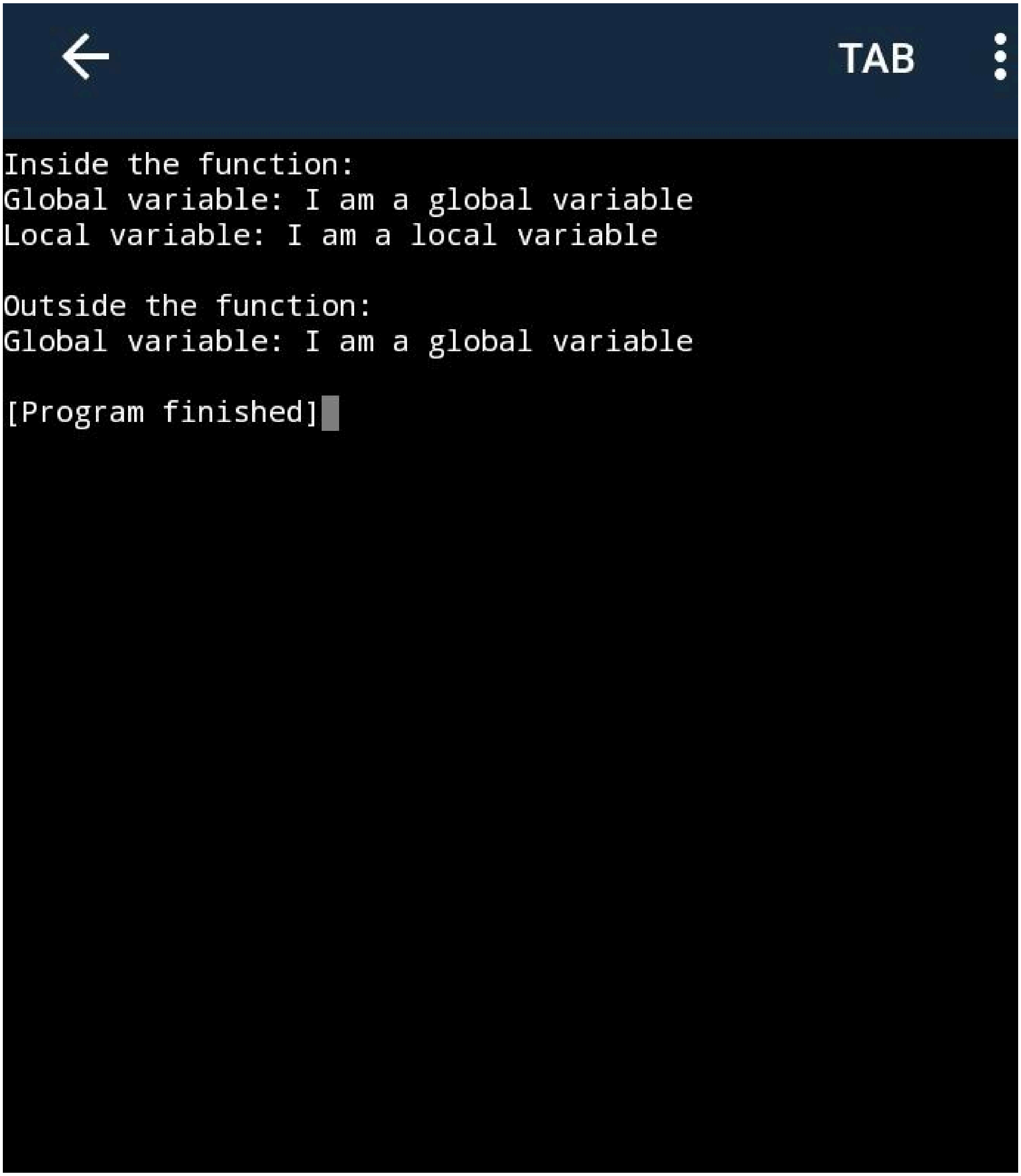
1. Aim:-
To demonstrate the use of local and global variables in Python.

2. Python Code:-

new* new* new* new* new*

```python
message = "I am a global variable"

def display_message():
    # Local variable
    local_message = "I am a local variable"

    print("Inside the function:")
    print("Global variable:", message)
    print("Local variable:", local_message)


display_message()

print("\nOutside the function:")
print("Global variable:", message)
```

Output:-



4. Explanation:-

A global variable is declared outside a function and can be accessed throughout the program. A local variable is declared inside a function and can be accessed only within that function.

TASK 5 – Modules:-

1. Aim:-

To demonstrate the use of built-in Python modules such as math, random, and datetime.

2. Python Code:-

new* new* new* new* new*

```python
import math
import random
import datetime

# Using math module
number = 25
print("Square root of", number, ":", math.sqrt(number))

# Using random module
print("Random number between 1 and 100:", random.randint(1, 100))

# Using datetime module
current_date = datetime.date.today()
print("Today's date:", current_date)
```

Output:-

TAB

```
Square root of 25 : 5.0
Random number between 1 and 100: 78
Today's date: 2026-08-09

[Program finished]
```

Explanation:
This program demonstrates three built-in Python modules. The math module is used to calculate the square root, the random module generates a random number, and the datetime module displays the current date.